

Model formulation

Lecture notes for Modelling 2025

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See also:
Chapter 3 in Rasmuson

Today you should aim to:

1. Get familiar with the basic steps for formulating a model
2. Practice working with conservation, constitutive, and linkage equations
3. Start formulating some simple models

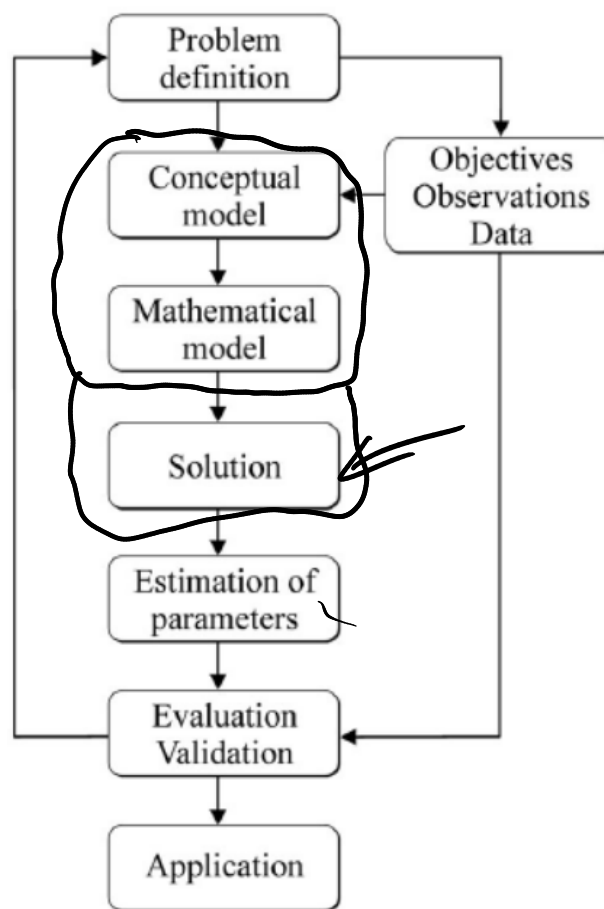


Figure 1.5. The different steps in model development.

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Start with warm-up problem 1 on heat production.

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Model formulation steps

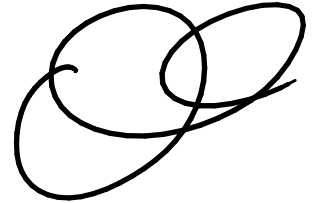
Formulating a model includes several steps that can be organized like this:

1. Initial concepts and assumptions (boundary, state variables, sketches)
2. Balance/conservation equations
3. Constitutive equations (transfer or rate equations)
4. Linking equations
5. Spatial or other simplifications
6. Initial or boundary conditions
7. Governing equation
8. Analytical or numerical solution

We'll go through some of these steps today.

These steps do not always apply to every problem.

And the list is somewhat arbitrary--there are alternative approaches.

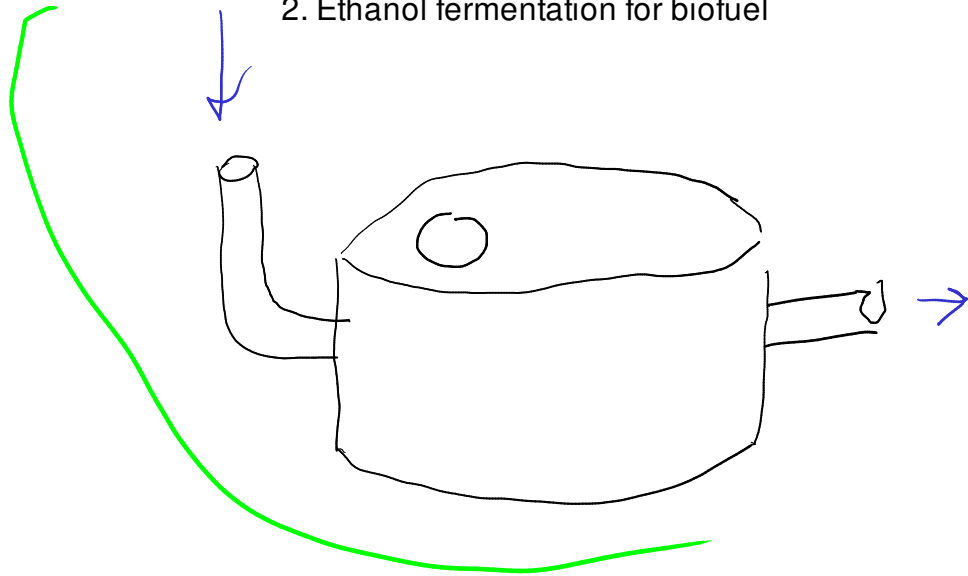


Example systems for modeling

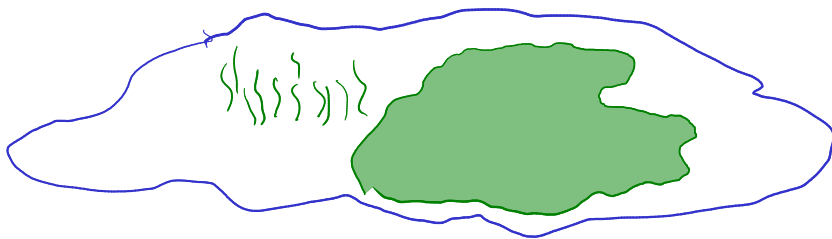
1. A cooling cup of coffee



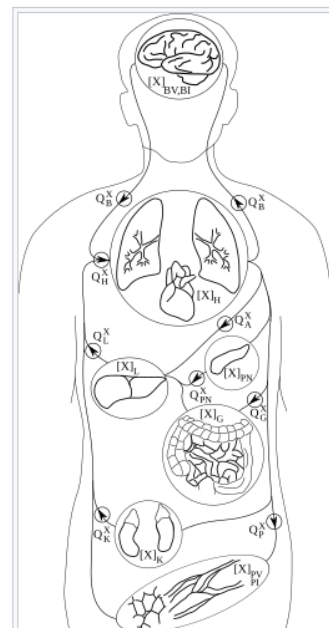
2. Ethanol fermentation for biofuel



3. A lake with water pollution, aquatic weeds, and toxic algae



4. Pharmacokinetic model for a drug



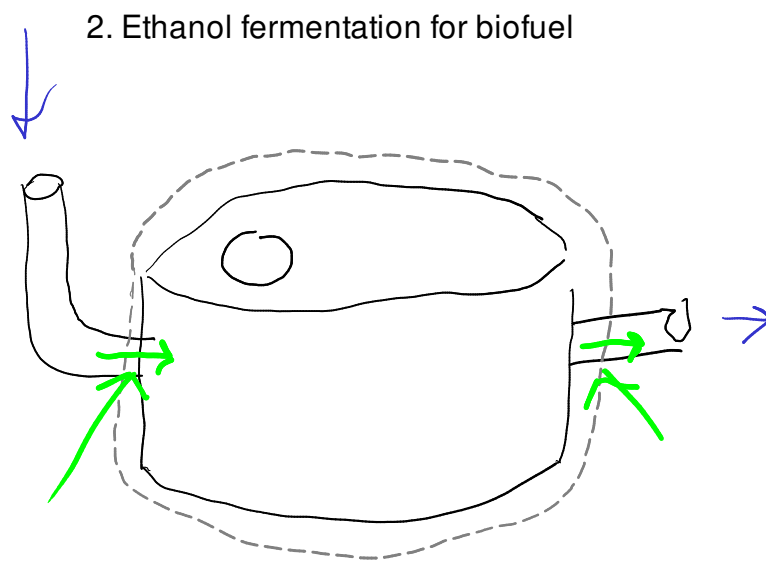
Graphic representation of a physiologically based whole body model. Here, it is dissected into seven tissue/organ compartments: brain, lungs and heart, pancreas, liver, gut, kidney and adipose/muscle tissue. Blood flows, Q , and concentration, $[X]$, of a substance of interest are depicted.

Model boundary

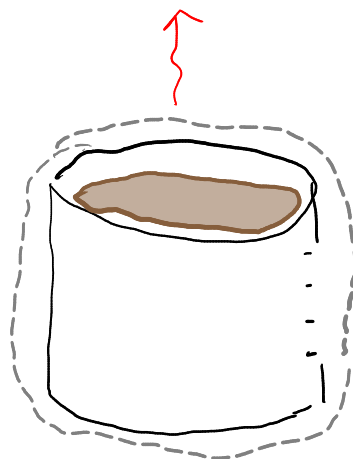
An explicit boundary facilitates development of model equations.

The boundary is often obvious.

- * Surround the volume of interest
- * Typically not interested in fate of material or energy outside boundary
- * May be able to measure or control material or energy flow through the boundary



1. A cooling cup of coffee



State variables

State variables are those that you are trying to model, often over time, that describe the state of the system of interest.

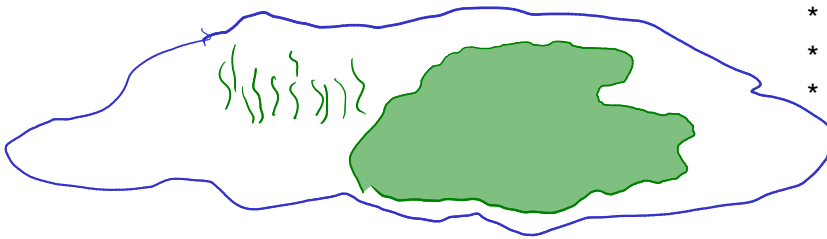
Warning: usage of the term "state variables" varies within discipline or source

1. A cooling cup of coffee



State variable: thermal energy

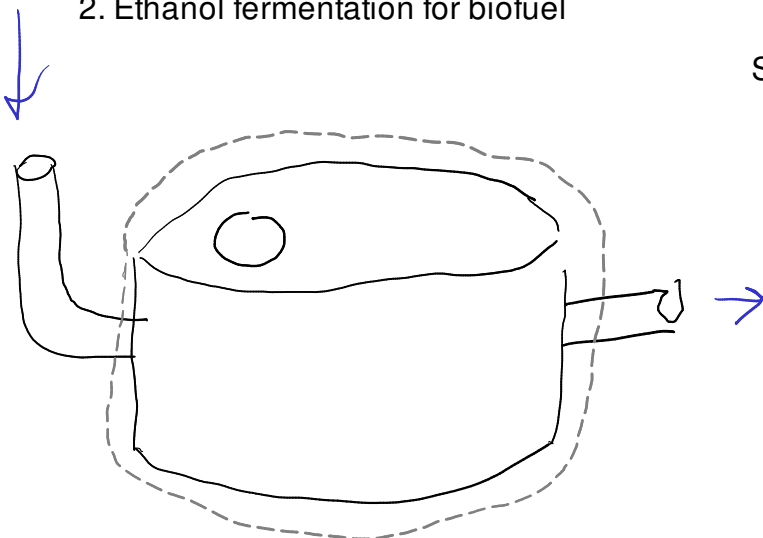
3. A lake with water pollution, aquatic weeds, and toxic algae



State variables:

- * Aquatic weed mass
- * Toxic algae mass
- * Nutrient mass (total N, total P perhaps)
- * Dissolved oxygen

2. Ethanol fermentation for biofuel



State variables?

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Balance or conservation equations

Think about mass (material), energy, species balance

General balance equation is:

$$\text{accumulation} = \text{transport} + \text{generation}$$

*typically all
total flow*

These rates might be:

g/s

g/s

$\uparrow T^{-1}$

W

And they might be normalized by volume or area:

But often afterwards

$$g \cdot m^{-2} \cdot s^{-1}$$

$$M L^{-2} T^{-1}$$

*flux
mass
flux*

$$J \cdot m^{-2} \cdot s^{-1} = W \cdot m^{-2}$$

heat flux

A model may include multiple balance equations

1. Our coffee example

$$\text{Rate of energy accumulation} = \text{Heat transfer in} - \text{Heat loss} + \text{Heat generation}$$

$$W \equiv J/s$$

2. Fermenter for ethanol production

$$\text{Rate of substrate accumulation inside fermenter} = \text{Flow of substrate in} - \text{Flow of substrate out} - \text{Substrate consumption}$$

3. Lake model

$$\text{Rate of inorganic N accumulation} = \text{Rate of N flowing in} - \text{Rate of N flowing out} + \text{N release by decomposition} - \text{N uptake by producers}$$

Constitutive relations (equations) - transport and other rate equations

Heat and mass transfer (transport) are described by parallel rate equations with the general form:

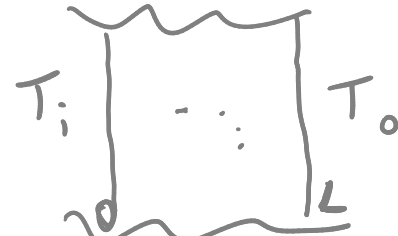
flux = transport coefficient x gradient

These include:

1. Fourier's law for heat transfer, e.g., steady state in a plane wall:

heat flux W/m^2 $q = k \Delta T / \Delta x$

thermal conductivity



2. Newton's law of cooling for convective heat loss:

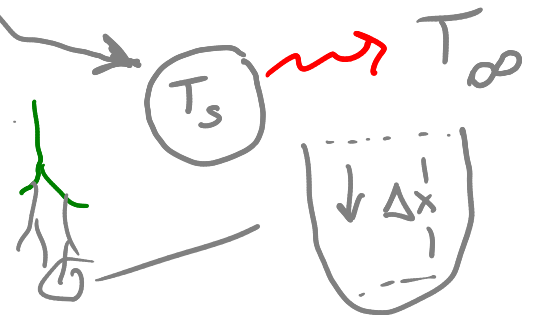
heat transfer coefficient $q = h(T_s - T_\infty)$

$\Delta T = T_i - T_o$

3. Mass diffusion, Fick's law, e.g., steady state in a plane wall

glucose mass flux $g/m^2 \cdot s$ $j = D \cdot \frac{\Delta c}{\Delta x}$

diffusion coeff.



In biological, completely empirical relationships are often used.

A first-order equation is common in chemistry and physics and often used in empirical models

4. First-order reaction

Reaction rate g/s $r = k \cdot m$ *1st order rate constant* *mass of reactant*

$g/m^2 \cdot s \rightarrow r = k \cdot c$ *concentration*

Transport law
general form:
flux = constant · gradient

Example 2?

See Table 3.1 in Rasmuson

* State variables
* Material properties

Other variables and "linkage equations"

We typically have more than just state variables in a model.

Other variables may be needed to close the balances, to determine rates, or otherwise may be of interest.

These others might be linked to the state variable, or may need to be added as additional state variables.

Coffee example:

The rate of thermal energy transfer, or heat transfer, depends on the temperature of the coffee. And anyway, it is temperature that we can measure.

So we should include temperature in our model as well.

Specific heat and density link temperature to thermal energy.

Microbial growth:

Microbial biomass production is linked to substrate consumption ("utilization rate") through a yield Y .

Oxygen demand is typically used for units

$$W \equiv J/s$$
$$q = h \cdot (T_s - T_{air})$$

$$\frac{dX_a}{dt} = Y \frac{dS}{dt}$$

Mass balance complications

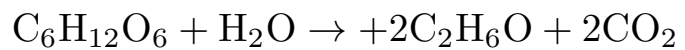
Matter can change forms and a total mass balance is often not that useful.

Instead it usually makes sense to focus on some specific component:

- * Mass of a particular compound,
- * mass of a particular element,
- * chemical oxygen demand, ...

Stoichiometry is very useful!

Ethanol fermentation example:



What are the stoichiometric coefficients for glucose to ethanol on a molar basis? [menti.com](https://www.menti.com)

Carbon mass basis? Compound mass basis? COD basis?

How would you write an equation that quantitatively links ethanol production to glucose consumption?

A linkage equation

Energy balance complications

For energy, it is actually total energy that is conserved (ignoring nuclear reactions): kinetic, potential, chemical, thermal (also kinetic, but thermal for us).

But we can often ignore all but thermal when dealing with heat transfer.

It can be convenient to use fixed generation term instead of complete energy balance.

Self-heating of anaerobic digesters using energy crops

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Abstract With the increasing application of energy crops in agricultural biogas plants and increasing digester volumes, the phenomenon of self-heating in anaerobic digesters appeared in some cases. Until now this development was just known from aerobic systems. To obtain an idea of the thermodynamics inside an anaerobic digester, a detailed analysis of all heat fluxes in a full-scale agricultural biogas plant was carried out. Several experiments were realised to quantify the influences of different internal and external energy sources. To estimate the impact of self-heating in anaerobic systems, data of other full-scale agricultural biogas plants in Austria were collected. Alternatives to the cooling of the digesters are discussed based on individual experiences of several plants. A connection between carbohydrate-rich substrates, especially with high starch contents, and the self-heating could be shown. From the results it can be assumed that the anaerobic digestion of most energy crops is exothermic, which is in contrast to the current thermodynamic belief.

Keywords Anaerobic digestion; digester thermodynamics; energy crops; heat fluxes; self-heating

$$\begin{array}{ccccccc} \text{Rate of thermal} & & \text{Heat transfer} & & \text{Heat} & & \text{Heat} \\ \text{energy} & & \text{in} & & \text{loss} & & \text{generation} \\ \text{accumulation} & = & & - & & + & \\ & & & & & & \end{array}$$

Think about the human heat generation problem.

A linkage equation

Relating heat and temperature

Heat accumulation cannot be measured directly in J.

Temperature is measured and the two are linked by specific heat capacity.

Thermal energy change

=

$$\Delta Q = \Delta T \cdot c_p \cdot m$$

Handwritten notes:
 ΔQ is circled with "J" above it.
 ΔT has " $^{\circ}\text{C} = \text{K}$ " above it.
 c_p has "specific heat" written above it.
 m has "g" written next to it.
 Below the equation, there is a handwritten unit: $\text{J} / \text{K} \cdot \text{g}$.

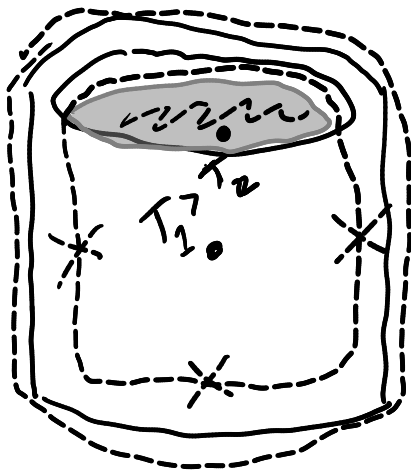
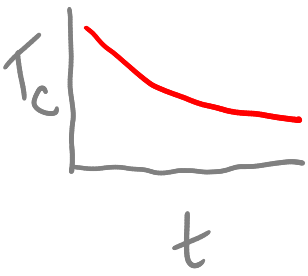
Reference temperature, change in thermal energy

Typical units for specific heat: kJ/kg-K

Substance	Phase	Isobaric mass heat capacity c_p $\text{J} \cdot \text{g}^{-1} \cdot \text{K}^{-1}$	Molar heat capacity, $C_{p,m}$ and $C_{v,m}$ $\text{J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$		Isobaric volumetric heat capacity $C_{p,v}$ $\text{J} \cdot \text{cm}^{-3} \cdot \text{K}^{-1}$	Isochoric molar by atom heat capacity $C_{v,am}$ $\text{mol} \cdot \text{atom}^{-1}$
			Isobaric	Isochoric		
Titanium	solid	0.523	26.060		2.638	3.13 R
Tungsten ^[3]	solid	0.134	24.800		2.580	2.98 R
Uranium	solid	0.116	27.700		2.216	3.33 R
Water at 100 °C (steam)	gas	2.030	36.500	27.500	1.530	
Water at 25 °C	liquid	4.181	75.340	74.550	4.138	
Water at 100 °C	liquid	4.216 <i>[dubious - discuss]</i>	75.950	67.900	3.770	
Water at -10 °C (ice) ^[3]	solid	2.050	38.090		1.938	
Zinc ^[3]	solid	0.387	25.200		2.760	3.03 R
Substance	Phase	Isobaric mass heat capacity c_p $\text{J} \cdot \text{g}^{-1} \cdot \text{K}^{-1}$	Isobaric molar heat capacity $C_{p,m}$ $\text{J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$	Isochore molar heat capacity $C_{v,m}$ $\text{J} \cdot \text{mol}^{-1} \cdot \text{K}^{-1}$	Isobaric volumetric heat capacity $C_{p,v}$ $\text{J} \cdot \text{cm}^{-3} \cdot \text{K}^{-1}$	Isochore atom-molar heat capacity in units of R $C_{v,am}$ $\text{atom} \cdot \text{mol}^{-1}$

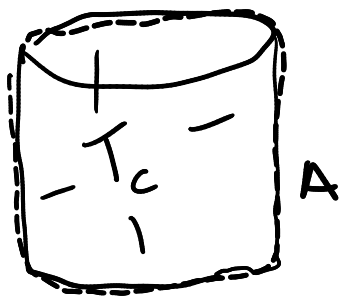
Example of a linkage equation

Coffee cooling model



$T \rightarrow Q$ thermal energy

$$q = h(T_s - T_{air})$$



Lumped capacitance (parameter) model

$$w/m^2 \quad q = h(T_c - T_{air})$$

$$w \quad \dot{Q} = q \cdot A$$

Boundary conditions

Boundary conditions (BCs) describe the model at the edges of its domain.

Relevant to partial differential equation (PDE) models - spatial models - 1D, 2D, 3D models.

BCs describe how a system interacts with surrounding space that is outside of the model boundary.

They link all the internal math, which usually describes how things change, to some fixed values that can be measured.

Boundary conditions can take many different forms.

Table 3.2. Common boundary conditions for use with the transport of mass.

Description	Math
Concentration at a boundary is specified	$C = C_0$
Mass flux across a boundary is continuous	$(n_i)_{x=0-} = (n_i)_{x=0+}$
Concentrations on both sides of a boundary are related functionally	$(C_i)_{x=0-} = f(C_i)_{x=0+}$
Convective mass (mole) flux at a boundary is specified	$(N_A)_{x=0} = k_c(C_{bulk} - C_{surface})$
Rate of reaction at a boundary is specified	$(N_A)_{x=0} = R_A$

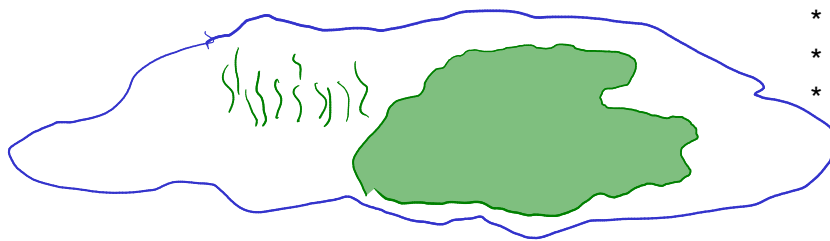
Initial conditions

Initial conditions (ICs) describe the initial state of a system - the values of state variables.

Maybe a part of both ODE and PDE models.

ICs can be used with other equations to solve for parameter values or programmed explicitly in a numerical solution.

3. A lake with water pollution, aquatic weeds, and toxic algae



State variables:

- * Aquatic weed mass
- * Toxic algae mass
- * Nutrient mass (total N, total P perhaps)
- * Dissolved oxygen

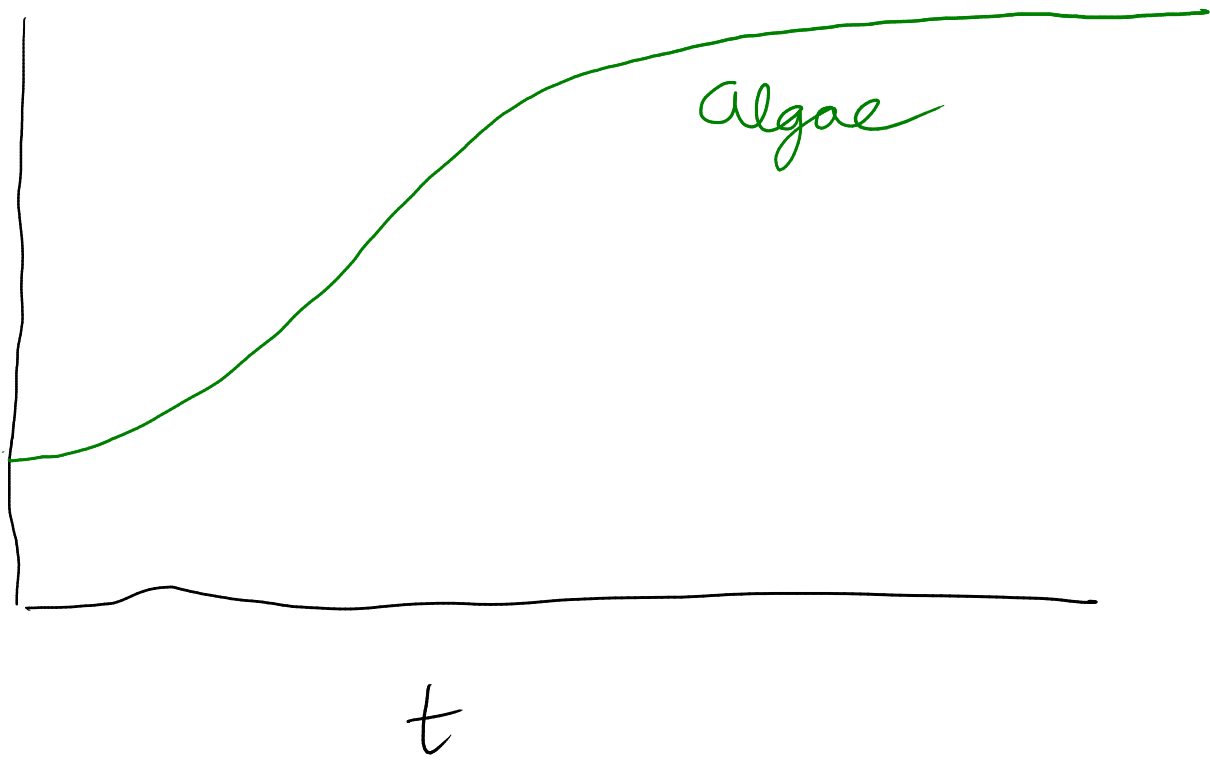
So ICs could be:

- * Initial weed = 30 t
- * Initial algae = 50 kg
- * Initial N = 400 kg
- * Initial P = 20 kg
- * Initial DO = 5 kg

Values for initial conditions affect dynamics and may or may not affect steady-state or equilibrium values, depending on the model.

Just like reality!

Take scenario 1 - high N & P



2. Low N & P



1. A cooling cup of coffee



Governing equations

A governing equation is the differential equation that describes how the state variables change over time and space.

It is developed by combining conservation, constitutive, and linkage equations.

Rasmuson has some examples for transport in Appendix A1, e.g.,

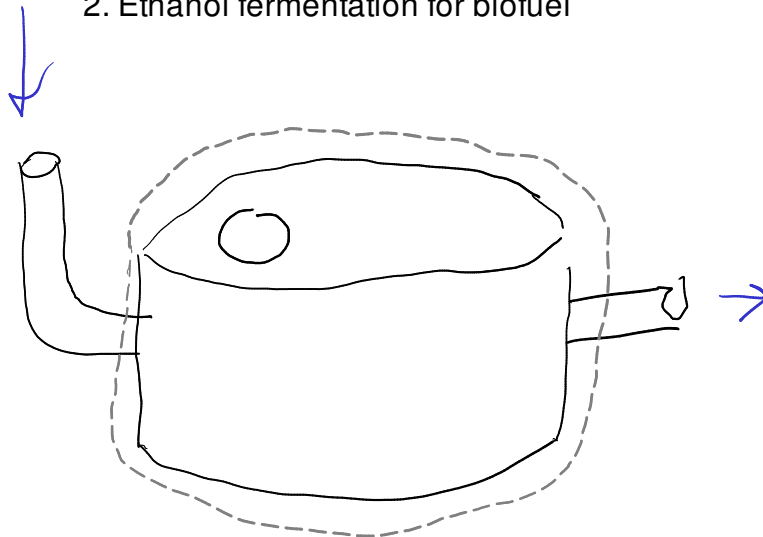
Mass balance for component A (ρ and D_{AB} constant)

$$\frac{\partial C_A}{\partial t} + v_x \frac{\partial C_A}{\partial x} + v_y \frac{\partial C_A}{\partial y} + v_z \frac{\partial C_A}{\partial z} = D_{AB} \left(\frac{\partial^2 C_A}{\partial x^2} + \frac{\partial^2 C_A}{\partial y^2} + \frac{\partial^2 C_A}{\partial z^2} \right) + R_A.$$

This and others can be derived by combining (mass) balance equations, constitutive equations, and possibly linkage equations

Existing governing equations can be used as a starting point, with terms eliminated based on the systems, e.g., y and z dimensions, advection, or steady state.

2. Ethanol fermentation for biofuel



Model solution

Solve the governing equation to get a model solution, which can then be applied with pencil and paper and a calculator or implemented in a computer program

Easy? Often not!

Analytical solutions

Numerical solutions

Model simplification

Why simplify? Because it could make:

- * analytical solution possible,
- * model easier to implement (program),
- * model easier to understand,
- * model easier to apply.

We need to consider:

- * Will simplification substantially change model error or uncertainty?
- * Can simplification lead to useful limiting cases?
- * Is the model still useful after simplifying?

How can models be simplified? Many different ways.

1. normalize variables and use dimensionless groups,
2. ignore spatial variation in state variables (lumped parameter),
3. solve for steady-state only,
4. use symmetry to reduce dimensions,
5. assume constant inputs,
6. ...

In some cases we have to simplify in order to evaluate the simplification.

Rates - residence time and turnover

Residence time
Retention time
Detention time
Turnover time
Turnover rate

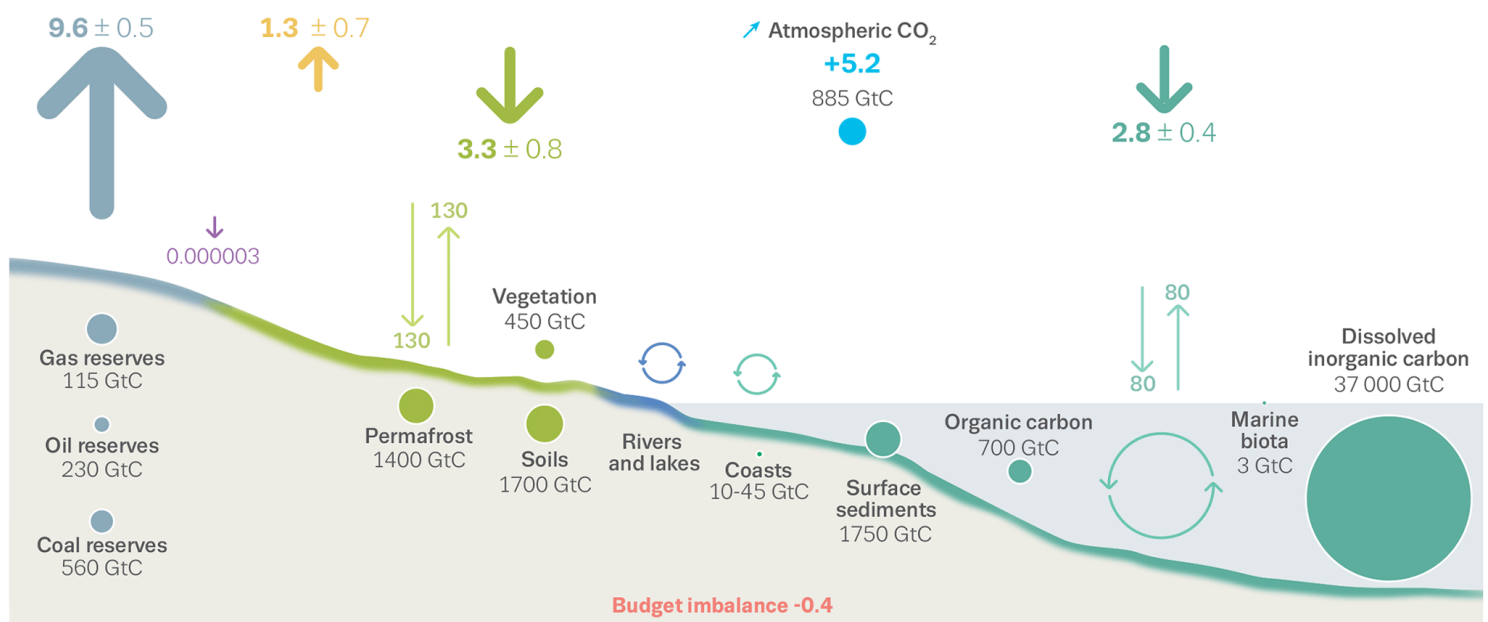
These are all different terms for a useful concept.

Can be useful as a normalized flow rate.

Residence time helps in predicting and understanding system behavior.

For example, what is the residence time of CO₂ in the atmosphere? Turnover rate?
How about oceans?

The global carbon cycle



Anthropogenic fluxes 2013-2022 average GtC per year

↑ Fossil CO₂ E_{FOS} ↓ Land uptake S_{LAND} ● Stocks GtC
↑ Land-use change E_{LUC} ↓ Ocean uptake S_{OCEAN} + Atmospheric increase G_{ATM}
↓ CDR not included in E_{LUC} ↑ Carbon cycling GtC per year ■ Budget Imbalance B_{IM}

<https://doi.org/10.5194/essd-15-5301-2023>